

Cheese Shelf-Life Studio — V7 Specialist Retrain Report

Six specialists (soft/semi-hard/hard × general/safety) × four algorithms (RF, LightGBM, XGBoost, EBM), trained on the corrected V7 datasets. Generated 2026-08-24 18:15 UTC. Parallel evaluation track — V6 artifacts and the live app are untouched.

Executive summary

- All three **general_shelf_life** specialists are excellent: test R^2 ranges 0.920–0.971.
- All three **safety_endpoint** specialists are degraded, in a pattern that tracks almost exactly with how many of their target values are right-censored ("at least N days") rather than exact — see §5.
- **Hard / Safety endpoint** has effectively collapsed to predicting the training mean (test R^2 =-0.011) — confirmed directly, not inferred (§6).
- A real preprocessing bug (% w/w concentration unit, 17,662 V7 rows) was found and fixed during verification, not a data change (§8).
- No physical-form combination in V7 can reach "Strong" support — a real, honestly-reported regression vs V6's data (§9).
- No correction multiplier, offset, clipping, or manual calibration was added anywhere in this work (§10).

1. What changed in V7

The user replaced the raw specialist training data with three "corrected" V7 files. Verified directly against the data (not assumed): V7 adds 13 columns beyond V6's 62, most notably genuine challenge-context fields (`is_challenge_test` , `initial_inoculum_log_cfu_g` , `growth_support`) and a target-censoring flag (`target_is_lower_bound`). Row counts are lower than V6 (V7 totals 34,000 rows across the three category files vs V6's ~35,650), and vocabulary is narrower (soft: 19 food_matrix values, semi-hard: 11, hard: 6), consistent with a more literature-anchored, "corrected" scope.

2. Architecture & constraints

Same six-specialist architecture as V6: one model family per (cheese_category, model_task), each a Random Forest / LightGBM / XGBoost / EBM ensemble, routed automatically. New context-scoped train/validation/test splits were built fresh from the V7 data (leakage-safe, grouped by `context_id` , seed=42). The production model for each specialist is selected purely by validation RMSE. Saved under a new `artifacts_v7/` directory; `artifacts_v6/` is untouched (confirmed via unchanged file modification times). **No dataset regeneration, rebalancing, relabeling, or augmentation was performed, and no correction multiplier, offset, clipping, or manual calibration was added to any prediction** — confirmed by grep across every file touched in this work, both before and after.

3. Headline comparison — production model per specialist

Row highlighted where test $R^2 < 0.5$. "%LB test" = share of the specialist's own test-split targets that are right-censored lower bounds (see §5) — not a model defect, a property of the target.

SPECIALIST	BEST MODEL	N TRAIN	N VAL	N TEST	VAL R^2	TEST R^2	VAL MAE	TEST MAE	VAL RMSE	TEST RMSE	TRAIN-VAL GAP	%LB TEST
Soft / General shelf life	LightGBM	6160	1320	1320	0.972	0.971	2.39	2.53	4.49	4.89	0.025	0%
Soft / Safety endpoint	XGBoost	3360	720	720	0.592	0.594	8.31	7.73	10.42	10.34	0.239	69%
Semi-hard / General shelf life	LightGBM	5040	1080	1080	0.931	0.920	7.35	8.00	12.82	13.39	0.059	0%
Semi-hard / Safety endpoint	XGBoost	2800	600	600	0.115	0.317	14.81	13.26	17.30	16.05	0.353	93%
Hard / General shelf life	LightGBM	4200	900	900	0.916	0.932	8.03	6.93	12.26	11.30	0.076	0%
Hard / Safety endpoint	XGBoost	2240	480	480	0.008	-0.011	17.00	17.80	19.87	20.98	0.038	100%

4. Full results — all 4 algorithms per specialist

★ marks the production model (lowest validation RMSE). All numbers read directly from each specialist's saved `metrics.json` — nothing recomputed for this report.

Soft / General shelf life

ALGORITHM	TRAIN R ²	VAL R ²	TEST R ²	VAL MAE (D)	TEST MAE (D)	VAL RMSE (D)	TEST RMSE (D)	TEST MAPE %	TRAIN-VAL GAP
Random Forest	0.997	0.955	0.956	2.84	2.95	5.67	5.99	12.2	0.043
LightGBM ★	0.997	0.972	0.971	2.39	2.53	4.49	4.89	10.4	0.025
XGBoost	0.999	0.969	0.968	2.38	2.55	4.65	5.11	10.5	0.029
EBM	0.985	0.944	0.944	3.89	3.88	6.32	6.79	25.3	0.041

Soft / Safety endpoint

ALGORITHM	TRAIN R ²	VAL R ²	TEST R ²	VAL MAE (D)	TEST MAE (D)	VAL RMSE (D)	TEST RMSE (D)	TEST MAPE %	TRAIN-VAL GAP
Random Forest	0.997	0.548	0.572	8.62	7.85	10.97	10.62	26.8	0.449
LightGBM	0.876	0.583	0.573	8.43	7.85	10.54	10.61	29.0	0.292
XGBoost ★	0.832	0.592	0.594	8.31	7.73	10.42	10.34	27.4	0.239
EBM	0.944	0.393	0.360	10.10	10.48	12.72	12.98	44.6	0.551

Semi-hard / General shelf life

ALGORITHM	TRAIN R ²	VAL R ²	TEST R ²	VAL MAE (D)	TEST MAE (D)	VAL RMSE (D)	TEST RMSE (D)	TEST MAPE %	TRAIN-VAL GAP
Random Forest	0.995	0.925	0.905	8.34	9.22	13.42	14.57	12.4	0.070
LightGBM ★	0.990	0.931	0.920	7.35	8.00	12.82	13.39	10.3	0.059
XGBoost	0.995	0.930	0.927	7.50	7.81	12.99	12.74	10.0	0.065
EBM	0.983	0.900	0.909	9.65	9.13	15.49	14.30	13.4	0.083

Semi-hard / Safety endpoint

ALGORITHM	TRAIN R ²	VAL R ²	TEST R ²	VAL MAE (D)	TEST MAE (D)	VAL RMSE (D)	TEST RMSE (D)	TEST MAPE %	TRAIN-VAL GAP
Random Forest	0.997	0.015	0.342	15.43	12.93	18.26	15.76	26.3	0.982
LightGBM	0.449	0.089	0.303	15.06	13.60	17.56	16.21	36.8	0.360
XGBoost ★	0.468	0.115	0.317	14.81	13.26	17.30	16.05	31.4	0.353
EBM	0.936	-0.280	-0.127	17.00	17.17	20.81	20.63	37.0	1.217

Hard / General shelf life

ALGORITHM	TRAIN R ²	VAL R ²	TEST R ²	VAL MAE (D)	TEST MAE (D)	VAL RMSE (D)	TEST RMSE (D)	TEST MAPE %	TRAIN-VAL GAP
Random Forest	0.994	0.910	0.927	8.48	7.82	12.68	11.68	9.7	0.084
LightGBM ★	0.991	0.916	0.932	8.03	6.93	12.26	11.30	8.3	0.076
XGBoost	0.995	0.911	0.933	8.05	6.96	12.55	11.26	8.4	0.083

ALGORITHM	TRAIN R ²	VAL R ²	TEST R ²	VAL MAE (D)	TEST MAE (D)	VAL RMSE (D)	TEST RMSE (D)	TEST MAPE %	TRAIN-VAL GAP
EBM	0.977	0.880	0.917	9.43	8.31	14.58	12.47	10.1	0.096

Hard / Safety endpoint

ALGORITHM	TRAIN R ²	VAL R ²	TEST R ²	VAL MAE (D)	TEST MAE (D)	VAL RMSE (D)	TEST RMSE (D)	TEST MAPE %	TRAIN-VAL GAP
Random Forest	0.996	-0.131	-0.097	17.96	18.28	21.22	21.86	22.9	1.127
LightGBM	0.036	-0.003	-0.024	17.14	17.90	19.98	21.12	22.4	0.039
XGBoost ★	0.046	0.008	-0.011	17.00	17.80	19.87	20.98	22.2	0.038
EBM	0.953	-0.731	-0.528	20.94	21.00	26.25	25.79	26.4	1.683

5. The censored-target finding

V7 introduces `target_is_lower_bound` : a flag on rows where the literature source reports "no spoilage/no growth by day N" rather than an exact shelf-life value — the true value could be materially higher than N. Standard regression treats every target as exact. The proportion of censored targets rises sharply from soft → semi-hard → hard safety specialists, and **test R² falls in the same order**:

SPECIALIST	%LB TRAIN	%LB VALIDATION	%LB TEST	TEST R ²
Soft / General shelf life	0%	0%	0%	0.971
Soft / Safety endpoint	73%	73%	69%	0.594
Semi-hard / General shelf life	0%	0%	0%	0.920
Semi-hard / Safety endpoint	95%	94%	93%	0.317
Hard / General shelf life	0%	0%	0%	0.932
Hard / Safety endpoint	100%	100%	100%	-0.011

This is a genuine data-learnability problem, not a preprocessing or wiring bug: a model trained to hit an exact-value loss on a target that is secretly "at least N" for most rows cannot help but be biased/underdetermined. It was **not** fixed with a correction here, per instruction — it is reported so the forthcoming independent V7 test set (or a future censoring-aware retrain, e.g. Tobit/survival-style loss) can address it deliberately.

6. Deep dive: the collapsed hard/safety_endpoint specialist

Directly inspected (not inferred) by comparing the true vs. predicted training-target standard deviation for each safety specialist's production model:

SPECIALIST	MODEL	TRUE TARGET STD (D)	PREDICTED STD (D)	PREDICTED STD AS % OF TRUE
Soft / Safety endpoint	XGBoost	16.31	13.01	79.8%
Semi-hard / Safety endpoint	XGBoost	20.07	8.96	44.6%
Hard / Safety endpoint	XGBoost	21.06	0.77	3.6%

Hard/safety's XGBoost model predicts almost the same value (the training mean) for every row, regardless of input — consistent with its validation-loss early stopping firing after just 2 boosting rounds (see §4). This is the direct cause of its negative test R².

7. New V7 feature wiring — verified, not assumed

`initial_inoculum_log_cfu_g` and `growth_support` are populated only for safety_endpoint rows in the raw data. A generic rule in the training script (not a hardcoded "safety-only" special case) excludes any column that is 100% null within a specialist's own task-filtered training data. Spot-checked directly against the saved schemas:

CATEGORY	GENERAL: N FEATURES	GENERAL: CHALLENGE FIELDS	SAFETY: N FEATURES	SAFETY: CHALLENGE FIELDS
Soft	28	correctly absent	30	correctly present
Semi-hard	28	correctly absent	30	correctly present
Hard	28	correctly absent	30	correctly present

Live sensitivity check (re-run fresh for this report, not cached): for each safety specialist, one representative row was scored with every `growth_support` value the specialist was trained on, to confirm the feature actually moves the prediction rather than just existing in the schema:

SPECIALIST	BASELINE PREDICTION	PREDICTIONS BY GROWTH_SUPPORT	RESULT
Soft / Safety endpoint	21.5	growth: 21.5d, limited: 39.4d, no_growth: 39.6d	responsive
Semi-hard / Safety endpoint	27.1	growth: 27.1d, limited: 61.9d, no_growth: 59.7d	responsive
Hard / Safety endpoint	82.0	limited: 82.0d, no_growth: 82.0d	flat (no response)

Soft and semi-hard safety specialists respond measurably to `growth_support` (confirming the feature is live). Hard/safety's flat response is the same collapsed-model symptom from §6, not a feature-wiring failure specific to `growth_support/inoculum` — it doesn't respond to storage temperature either (checked separately).

8. Bug found and fixed during verification: % w/w concentration unit

All three V7 files use % w/w as a **raw** `primary_concentration_unit` spelling (17,662 rows total). `concentration_units.py` 's conversion table had `"%"` and `"wt %"` mapping to canonical % w/w, but no entry for the raw spelling % w/w itself — `to_canonical()` would have raised `UnrecognizedConcentrationUnit` on every one of those rows at prediction time. Verified empirically (module's own methodology): the canonical/raw value ratio for these rows is exactly 1.0 with zero variance across all 17,662 rows. Fixed by adding the verified self-mapped entry and extending the module's `--verify` check to cover V7 as well as V6. This is a preprocessing correctness fix, not a change to any training or test data.

9. Physical-form support regression (data property, reported not fixed)

In V7, `physical_form_observed` is 0 for every row in all three category files, and `physical_form_source` is always `"literature_supported_typical_form"` (never `"explicit_matrix_name"`, which some V6 rows had). Under the existing, unmodified support-level bands, this means **no V7 prediction can reach "Strong" physical-form support** — the best achievable is "Moderate" (confidence ≥ 0.85). This is a real characteristic of the V7 data, surfaced honestly through the existing support-assessment machinery, not a bug and not something this pass changed the bands to hide.

10. No prediction hacks — confirmed

Grepped for multiplier/offset/clip/correction/manual-calibration patterns across every file touched in this work, before and after implementation. No matches other than comments explicitly documenting the absence of such code. No dataset file under `data/raw/` was modified, regenerated, rebalanced, or relabeled at any point in this work.

11. Recommendations

1. Do not attempt to "fix" the safety specialists' R^2 with a correction factor — the root cause is target censoring, and a correction would hide rather than address it.
2. When the independent V7 external test set is built, consider excluding or separately reporting on rows whose true value is itself a lower bound, since a specialist trained on censored data cannot be fairly scored against an exact-value ground truth using standard R^2 /MAE.
3. If safety-endpoint accuracy needs to improve, the highest-leverage next step is a censoring-aware training objective (e.g. Tobit regression or a survival/hazard formulation) for the three safety specialists specifically — not more data volume or a different off-the-shelf algorithm, since all four algorithms show the same degradation pattern in §4.
4. General-shelf-life specialists need no further work — all three are strong and show only modest train-validation gaps.
5. Before any live promotion decision, re-run this evaluation against the user's forthcoming independent V7 external test set, not just V7's own held-out split.

Report generated from `artifacts_v7/*/metrics.json, training_manifest.json, schema.json, predictions.parquet`, produced by `train_specialists.py --data-version v7` against the V7 CSVs in `data/raw/`. Sensitivity and schema-check figures in §6–7 were computed fresh at report-generation time via `SpecialistRegistry(data_version="v7")`, not cached. No retraining occurred in the course of generating this report.